FOR A FLAT LOCAL COMPLETE INTERSECTION MORPHISM THE GYSIN CLASS IS THE FLAT PULLBACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FF4 (sources/stacks-chow/pinned-original-source.tex line 13547).

Allowed standard prerequisites (editorial).

- The flat pullback as a bivariant class. Same chapter, Lemma labelled lemma-flat-pullback-bivariant, invoked in the statement. Supplies the class that f^! is shown to equal.
- Testing identities of bivariant classes on an integral base. Same chapter, Lemma labelled lemma-bivariant-zero, cited at proof lines 13563-13565. Reduces the proof to f^*[Y] = i^!(g^*[Y]) with Y integral.
- Base change of smooth, syntomic and flat local complete intersection morphisms. Morphisms, Lemmas labelled morphisms-lemma-base-change-smooth and morphisms-lemma-base-change-syntomic, and More on Morphisms, Lemma labelled more-morphisms-lemma-flat-lci, cited at proof lines 13557-13561. Allows the reduction to an integral target.
- The Gysin class of a regular immersion on the fundamental class. Same chapter, Lemma labelled lemma-gysin-fundamental, cited at proof lines 13570-13572. Gives i^![P]_{n+t} = [X]_{n+s}.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B78: sources/stacks-chow/pinned-original-source.tex, source lines 6131–6131. Same-chapter pointer into the author environment lemma at source line 6131; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01VB: sources/stacks-chow/tags.txt, source lines 2440–2440. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13560 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01UI: sources/stacks-chow/tags.txt, source lines 2411–2411. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13561 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 069K: sources/stacks-chow/tags.txt, source lines 8044–8044. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13562 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02UC: sources/stacks-chow/pinned-original-source.tex, source lines 6585–6585. Same-chapter pointer into the author environment lemma at source line 6585; invoked at proof line 13564 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FBM: sources/stacks-chow/pinned-original-source.tex, source lines 12076–12076. Same-chapter pointer into the author environment lemma at source line 12076; invoked at proof line 13570 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let $f : X \to Y$ be a local complete intersection morphism of schemes locally of finite type over S. If the Gysin map exists for f and f is flat, then $f^!$ is equal to the bivariant class of Lemma 0B78.*

Proof. Choose a factorization $f = g \circ i$ with $i : X \to P$ an immersion and $g : P \to Y$ smooth. Observe that for any morphism $Y' \to Y$ which is locally of finite type, the base changes of f', g', i' satisfy the same assumptions (see Morphisms, Lemmas 01VB and 01UI and More on Morphisms, Lemma 069K). Thus we reduce to proving that $f^*[Y] = i^!(g^*[Y])$ in case Y is integral, see Lemma 02UC. Set $n = \dim_\delta(Y)$. After decomposing X and P into connected components we may assume f is flat of relative dimension r and g is smooth of relative dimension t. Then $f^*[Y] = [X]_{n+s}$ and $g^*[Y] = [P]_{n+t}$. On the other hand i is a regular immersion of codimension $t - s$. Thus $i^![P]_{n+t} = [X]_{n+s}$ (Lemma 0FBM) and the proof is complete. □